

Unit 11, Lesson 9: Revisiting Relationships in Circles

Benchmarks

MA.7.GR.1.3 Explore the proportional relationship between circumferences and diameters of circles. Apply a formula for the circumference of a circle to solve mathematical and real-world problems.

MA.7.GR.1.4 Explore and apply a formula to find the area of a circle to solve mathematical and real-world problems.

Learning Targets

- ☐ I can find the circumference of a circle in terms of π .
- ☐ I can find the area of a circle in terms of π .
- ☐ I can relate proportional relationships to circles.

11.9.1 Warm-Up: Bell Work

A series of screenshots of Circle M are shown. In each screenshot, the line segment and arc that are the same color have the same length.

Description	Screenshot
The diameter, KE , is wrapped around the circle starting at point E and ending at point D .	
The diameter, KE , is wrapped around the circle starting at point D and ending at point F .	
The diameter, KE , is wrapped around the circle starting at point F and ending at point H .	
A small part of Circle M 's circumference is shown between points H and E . Its equivalent length is shown on the diameter.	
All 3 diameters are shown wrapped around Circle M 's circumference, as well as the small amount of the circumference that is not a complete diameter.	

1. Complete the statements to describe the relationship shown in the series of screenshots of Circle M .

The

☐ perimeter
☐ circumference

 any circle is equal to the length of the

☐ radius
☐ diameter

times π . The value of π is approximately _____. The screenshots show the

length of the

☐ perimeter
☐ circumference

 is the same as the

length of slightly more than

☐ 2
☐ 3
☐ 4

☐ radii.
☐ diameters.

2. The part of the circumference that represents the small amount of the circumference that is not a complete diameter has an approximate length of _____.

11.9.2 Exploration: Revisiting Relationships in Circles

The formulas for finding the circumference and area of a circle are shown.



Area of a Circle, A
 $A = \pi r^2$, where r is the radius

Circumference of a Circle, C
 $C = \pi d$ or $C = 2\pi r$,
where d is the diameter and r is the radius

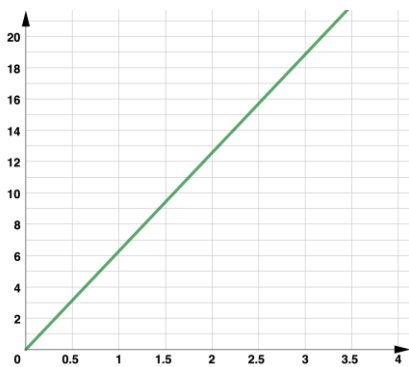
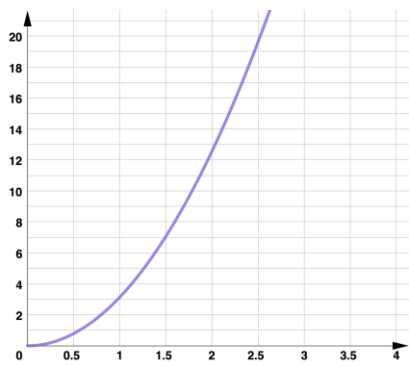
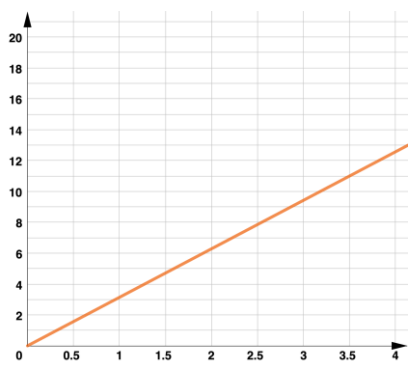
1. Work with your partner to complete each table of values in terms of π .

Table A					
Diameter (inches)	1	2	3	4	5
Circumference (inches)					

Table B					
Radius (inches)	1	2	3	4	5
Circumference (inches)					

Table C					
Radius (inches)	1	2	3	4	5
Area (sq. inches)					

2.
- Three graphs are shown.
On your own, mark the table that matches each graph and label the x - and y -axes appropriately.

Graph	Table
	☐ Table A ☐ Table B ☐ Table C
	☐ Table A ☐ Table B ☐ Table C
	☐ Table A ☐ Table B ☐ Table C

3.
- Check your work with your partner and if necessary, resolve any differences.
4.
- Discuss with your partner which relationships appear to be proportional, including what evidence there is to support your thinking. Then, summarize your discussion.

11.9.3 Bring It Together: Revisiting Relationships in Circles

Complete the statements.

1. There ☐ is
☐ is not a proportional relationship between the diameter and circumference of a circle.

- ☐ Therefore, there is no constant of proportionality.
- ☐ The constant of proportionality is _____.

2. There ☐ is
☐ is not a proportional relationship between the radius and circumference of a circle.

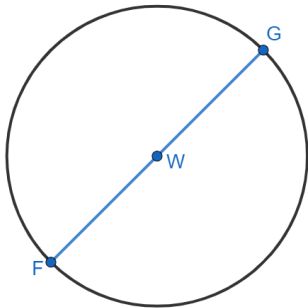
- ☐ Therefore, there is no constant of proportionality.
- ☐ The constant of proportionality is _____.

3. There ☐ is
☐ is not a proportional relationship between the radius and area of a circle.

- ☐ Therefore, there is no constant of proportionality.
- ☐ The constant of proportionality is _____.

11.9.4 Collaboration: Diameter and Circumference

Circle W is shown with diameter FG .



1. Miranda uses a drawing program to change the size of Circle W . Find the circumference, in terms of π , for each diameter she uses.

Diameter	Circumference
12	
19	
45	
146	

2. Suppose x represents the diameter and y represents the circumference. Write the constant of proportionality, $\frac{y}{x}$.
3. Complete the statement.
As the value of the diameter increases by 1, the value of circumference increases by _____.
4. Write an equation that represents the relationship between x , the diameter, and y , the circumference.

11.9.5 Wrap-Up: Reflection

1. Take a **SELFIE** by checking the box that best matches what you did today.

S howed my work	☐ 	☐ 
E xplained my answers	☐ 	☐ 
L ots of math vocabulary used	☐ 	☐ 
F ound connections to other content	☐ 	☐ 
I persevered through the problem	☐ 	☐ 
E liminated careless errors	☐ 	☐ 